



Write React

with TypeScript



Creating React/TypeScript project:

TypeScript provides [static typing for JavaScript](#), enhancing the development experience and [catching errors early](#).

Use [create-react-app](#) with the TypeScript template to set up a React project with TypeScript support.

Basic Code:



```
npx create-react-app my-app --template  
typescript
```



Adding return types for components:

In TypeScript, we can specify the **return type of a component** using `JSX.Element`

This helps TypeScript understand the **component's return value as JSX**, enabling type checking within the component.



```
const MyComponent: React.FC = ():  
  JSX.Element => {  
    // Component code here  
  };
```



Adding types to props:

TypeScript allows us to define types for props using interfaces.

Create an interface to **describe the expected props** and use it when declaring the component.



```
interface MyComponentProps {  
  name: string;  
  age: number;  
}  
  
const MyComponent:  
React.FC<MyComponentProps> = ({ name, age  
}) => {  
  // Component code here  
};
```



Adding types to events and event handlers:

Specify the event type in TypeScript to ensure proper typing for event handlers.

Use `React.MouseEvent` for mouse events or `React.ChangeEvent` for input change events.



```
interface ButtonProps {  
  onClick: (event:  
    React.MouseEvent<HTMLButtonElement>) =>  
    void;  
}  
  
const Button: React.FC<ButtonProps> = ({  
  onClick }) => {  
  // Component code here  
};
```





Adding types to hooks:

Provide **type annotations** when using hooks to ensure correct usage and type safety.

For example, use `useState<number>(0)` to indicate the initial state as a number.



```
const [count, setCount] = useState<number>(0);
```



Adding types to default props:

In TypeScript, you can define default props for components using the `defaultProps` property.

Specify the default values in the component's interface or directly in the component declaration

```
interface MyComponentProps {  
  name?: string;  
  age?: number;  
}  
  
MyComponent.defaultProps = {  
  name: 'John Doe',  
  age: 25,  
};
```

